

Marine Mental Strength (M2S) AI Use-Case for Mental Health Triage and Referral

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Introduction

The current stigma surrounding seeking help for mental health concerns remains a problem across all branches of the military. This report explains legal regulations, clinical and ethical considerations, technical design, and the validation process for Marine Mental Strength (M2S). M2S is a mental health and wellness online application aimed at reducing the stigma that prevents veterans and active-duty service members from seeking help. M2S enables Marines to chat with a state-of-the-art mental health large language model (LLM) to triage the individual and, if necessary, generate a referral to the nearest TRICARE-covered psychologist. The application will be hosted on the existing MHS GENESIS Patient Portal, which already allows Navy and Marine users to view health records and manage appointments. Application testing will be distributed to a small percentage of users, using site analytics to allocate most access to higher-activity users and a minority to lower-activity users.

Background

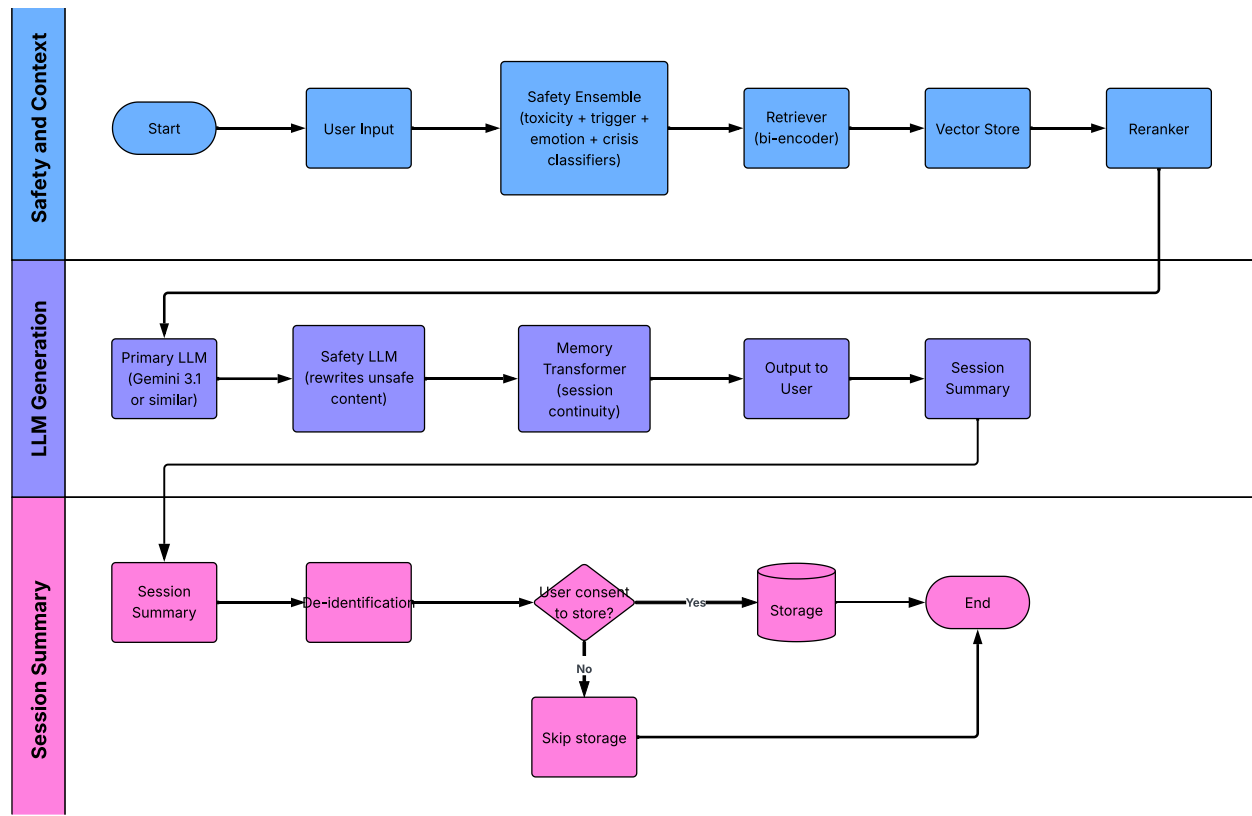
Many Marines report concerns such as “I don’t want to look weak” and “I don’t want to be unfit for service,” which contribute to the underutilization of mental health services. M2S is designed to provide a private, accessible entry point for service members to assess mental health concerns and receive guidance. The system leverages retrieval-augmented generation (RAG) to ground conversational outputs in curated domain knowledge and to reduce hallucinations by the generative model. The RAG core will include peer-reviewed literature, textbooks, and ethically acquired published works to provide context and improve the quality of responses.

Methodology

The original model will use a vetted pre-trained dataset procured via Hugging Face, supplemented by peer-reviewed, non-biased articles and publications focused on safe de-escalation and crisis recommendations. Published copyrighted works will be vetted and acquired through the author or controlling entity before inclusion in the RAG. Licensed pre-trained models (e.g., Ji et al., 2022) trained on consented social forum data about mental health will be verified for PII/PHI removal prior to incorporation.

The model architecture utilizes RAG with domain-curated knowledge, generative LLM layering, and transformer memory expansion implemented in Python 3.9 and PyTorch. The RAG uses a dual-encoder bi-transformer retriever to generate concurrent vector embeddings for comparison and improved retrieval. A metadata filter prevents unwanted vector selections during generation, and a reranker retrieves safe and relevant references for grounding. This augmented prompt, context, and grounding data are fed to the primary LLM (Gemini 2.5 Pro 27B) for supportive, non-clinical conversation. A secondary, smaller transformer (MentalBERT) acts as a safety governor to rewrite or block dangerous or misleading outputs prior to release. At session close, Gemini generates a high-level session summary. If the user consented to data use, the chat is processed by a transformer-based NER scrubber to de-identify text and replace PII/PHI with synthetic data; if the user did not consent, the partitioned data will be deleted and the partition destroyed. The application inherits PKI-based Authentication, Authorization, and Accounting (AAA) from the MHS GENESIS portal.

Figure 1: M2S System Data and Information Architecture



Legal and Ethical Considerations

M2S qualifies as a covered entity because it provides administrative electronic transactions and handles protected health information (PHI). The system follows the HIPAA Privacy Rule's de-identification standards using the Safe Harbor method, which removes 18 specified identifiers such as names, geographic subdivisions smaller than a state, and Social Security numbers (Health and Human Services, 2025). Pre-training includes scanning the data warehouse for corrupted, missing, or incorrect data. We inventory attributes containing PHI within the 18 identifiers and remove them using parameters designed to preserve model utility while protecting privacy. Removed PHI is replaced with synthetic data to help the model maintain contextual coherence across records. New users may opt in to donate their anonymized chat data to improve the system; donated data undergoes the same de-identification process.

The M2S landing page will require acceptance of terms of service and a privacy notice via a pop-up. The notice will briefly explain data use, PHI protection during chat, that PHI will not be used after the chat is closed, and that reproducing chat contents is discouraged because it could expose PII/PHI. The notice will also state that LLM outputs are not official diagnoses, that the model may hallucinate, and that unsafe or unwanted outputs should be reported before ending the chat. Data privacy attorneys will draft a Privacy Impact Assessment (PIA) for review by the Staff Judge Advocate (SJA) and the Chief Data and Artificial Intelligence Office (CDAO), followed by approval from the Department of the Navy (DoN) Chief Information Officer (CIO). Failure to meet these standards risks rejection or cancellation of the project and potential legal or administrative consequences.

Liability follows a hierarchical chain of command, beginning with the DoN CIO and cascading through the CDAO to the development team. After an incident, legal, administrative, or technical corrective actions will be taken based on root-cause analysis. Because tracing AI decision processes can be difficult, the terms of service and privacy notice explicitly warn users about hallucinations and limitations. M2S will apply the NIST AI Risk Management Framework (AI RMF) to analyze capabilities, predict edge cases, and mitigate dangerous outputs.

M2S will adapt to evolving federal regulations, including the December 11, 2025 executive order directing federal alignment on AI data practices. As HIPAA-related training data requirements change, models may need to adjust database schemas, attribute selection for training, and synthetic data replacement strategies. For example, if synthetic data proportion increases, model accuracy may change; this effect should be empirically tested rather than assumed. Current retention policies require authorized, non-synthetic data used for AI training to be retained for six years, after which it must be destroyed, anonymized, or deleted. If a patient opts out, their data

must be anonymized or deleted immediately. Active models will prevent access to de-identified patient nodes during operation, and all anonymization will follow the HIPAA Safe Harbor method (Health and Human Services, 2025).

Technical Challenges

The system must manage trade-offs between privacy and model performance when replacing PHI with synthetic data. Maintaining model accuracy while increasing synthetic data requires careful synthetic data generation and validation. The RAG pipeline must prevent retrieval of sensitive or irrelevant vectors through robust metadata filtering and reranking. Secondary safety models must reliably detect crisis language and dangerous instructions; false negatives or false positives in this layer present significant risk.

Integration with MHS GENESIS requires strict adherence to PKI authentication and the portal's AAA controls. Ensuring secure session handling, token management, and partitioned data deletion for non-consenting users are operational challenges. The system also faces the general limitations of LLMs, including hallucinations and unpredictable outputs, which necessitate layered safety checks and human oversight during validation and deployment.

Criticism Anticipation

M2S is not intended to replace clinical care. The tool is designed to initiate help-seeking and to identify needs prior to referral, not to provide definitive diagnoses. Users will be warned that the tool is for triage only. To reduce facility wait times, referrals include a pre-read summary for specialists. The RAG core will include books such as *Create Your Own Light: Finding Post-Traumatic Purpose* by Howze to provide relatable context and reduce the amount of user input required for personalized responses. Studies (e.g., Alanezi, 2023) indicate that users

supplementing in-person care with generative AI found value in psychoeducation, emotional support, self-assessment, and monitoring; however, reliance on AI for therapeutic interventions remains limited and should be supervised by clinicians.

Red-teaming and Institutional Review Board (IRB) collaboration will be used to identify exploitations and vulnerabilities prior to deployment. FDA Class II clearance is anticipated given the moderate-risk profile and potential for harm without safeguards; after FDA approval, the MHS Digital Policy Council will review the service to prevent redundant development and ensure governance.

Future Direction

M2S will iterate based on trial results, red-team findings, and regulatory changes. Continued tuning of synthetic data generation, retriever/reranker performance, and safety governor thresholds will be necessary to maintain accuracy and safety. Transparency about application limitations, known issues, and FAQs will be provided at the appropriate classification level. The long-term goal is to reduce stigma and improve access to mental health care for Marines by providing a safe, private, and effective triage pathway that integrates with existing TRICARE-covered services.

Conclusion

M2S aims to strengthen mental health and wellness among veteran and active-duty Marines by combining neural network technology, RAG grounding, and layered safety controls. With appropriate information security measures and strict policies to limit persistent data to only what is necessary, M2S connects users to an LLM for contextual conversations and referrals

while minimizing privacy risks. The system leverages MHS GENESIS authentication and authorization to ensure secure access and accountability.

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